Experiment Number: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA:

Resource Allocation, User Capacity and Mobile-Generation Context

Aim

To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and

CDMA, and examine how their assumed user capacity and interference behaviour differ.

Objectives

By the end of the experiment, students should be able to:

1. Model how FDMA, TDMA and CDMA allocate communication resources among

users.

2. Estimate and compare user capacity under stated assumptions.

3. Plot how estimated CDMA SIR changes as the number of simultaneous users increases.

4. Relate the access methods to representative mobile generations and interpret the limits

of the simplified model.

Requirements

• Computer with MATLAB

• No hardware is required

Theoretical Background

FDMA gives users separate frequency channels. TDMA gives users separate time slots. CDMA

allows users to share a frequency band and time by using different spreading codes.

This experiment uses a simplified analytical model to illustrate those differences. It assumes

ideal FDMA and TDMA resource separation, and estimates CDMA SIR using:

$SIR =$

G

$K - 1$

where G is the processing gain and K is the number of simultaneous users. SIR in decibels is:

$SIRdb = 10 \log 10\, SIR$

The model assumes ideal separation for FDMA and TDMA, and equal received power for CDMA users. It does not model fading, power control, adjacent-channel leakage, guard intervals or standard-specific constraints.

Procedure

Parameter	Value used	Meaning
Total bandwidth	1 MHz	Total available frequency bandwidth
FDMA channel width	200 kHz	Bandwidth allocated to each FDMA user
TDMA slots	8	Slots available per frame
CDMA processing gain	64	Assumed spreading/processing gain
CDMA target SIR	7 dB	Minimum assumed SIR for the capacity estimate

Algorithm:

a. Clear the MATLAB workspace, Command Window and open figures.

b. Enter the system parameters.

c. Estimate the FDMA and TDMA user limits.

d. Convert the CDMA target SIR from decibels to a linear value.

e. Estimate the CDMA user limit.

f. Calculate CDMA SIR for a range of simultaneous user counts.

g. Display the capacity table and generate the plots.

h. Compare the results with the assumptions and explain their limitations.

Procedure:

1. Run the MATLAB program below.

2. Record the assumed bandwidth, channel width, number of TDMA slots, CDMA processing gain and target SIR.

3. Inspect the printed capacity table and the capacity and CDMA-SIR plots.

4. Inspect the generation comparison table and the conceptual time-frequency resource

diagrams.

5. Change at least one parameter for each access method and rerun the program.

6. Compare the new and baseline results; explain the impact of the changed assumptions

Program 1 — Capacity estimate for FDMA, TDMA and CDMA

Objective 1: To estimate and compare the illustrative number of users supported by FDMA,

TDMA and CDMA using the specified system parameters.

MATLAB program:

```
>> % Comparison of FDMA, TDMA and CDMA

% Simplified illustrative model; not a full cellular-network simulation.

clear; clc; close all;

% Illustrative system parameters

totalBandwidthHz = 1e6; % Total bandwidth: 1 MHz

fdmaChannelHz = 200e3; % FDMA channel width: 200 kHz

tdmaSlots = 8; % Number of TDMA slots per frame

processingGain = 64; % CDMA processing gain

targetSIRdB = 7; % Minimum acceptable CDMA SIR

% Capacity estimates under the stated assumptions

fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);

tdmaUsers = tdmaSlots;

targetSIR = 10^(targetSIRdB / 10);

cdmaUsers = floor(1 + processingGain / targetSIR);

fprintf('Illustrative maximum supported users\n');

fprintf('FDMA: %d users\n', fdmaUsers);

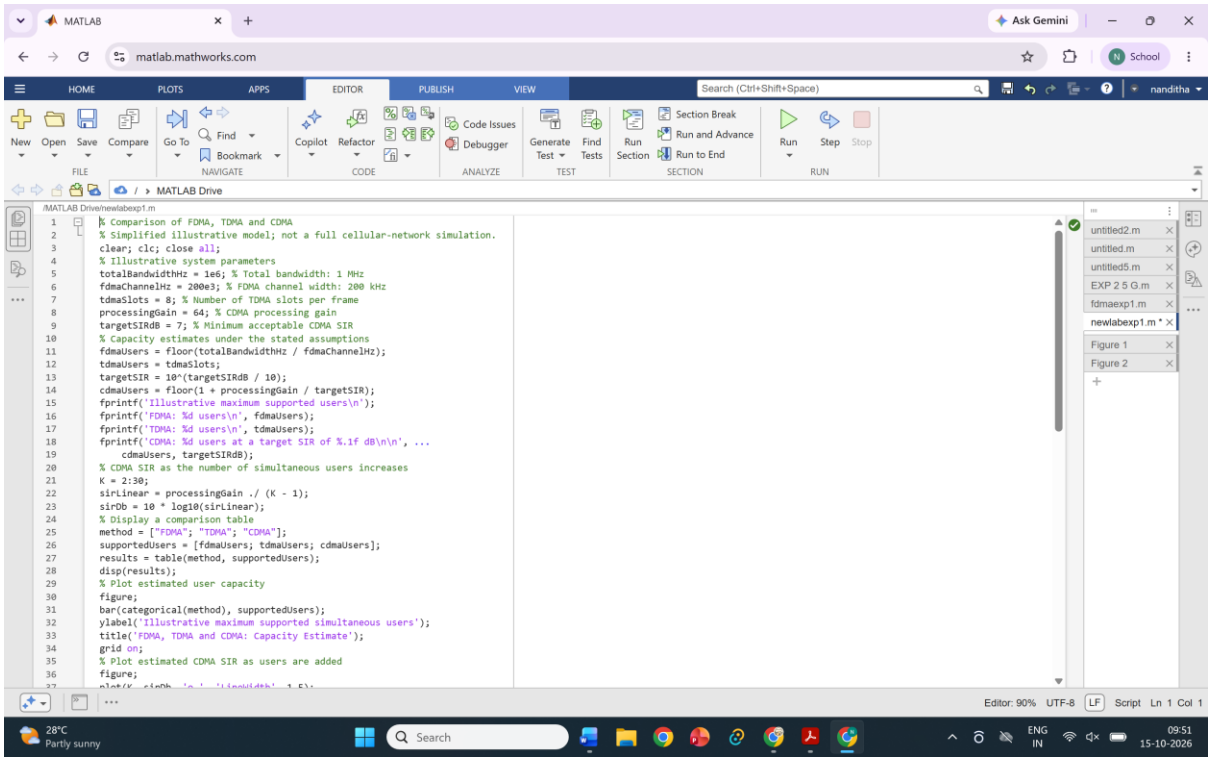
fprintf('TDMA: %d users\n', tdmaUsers);

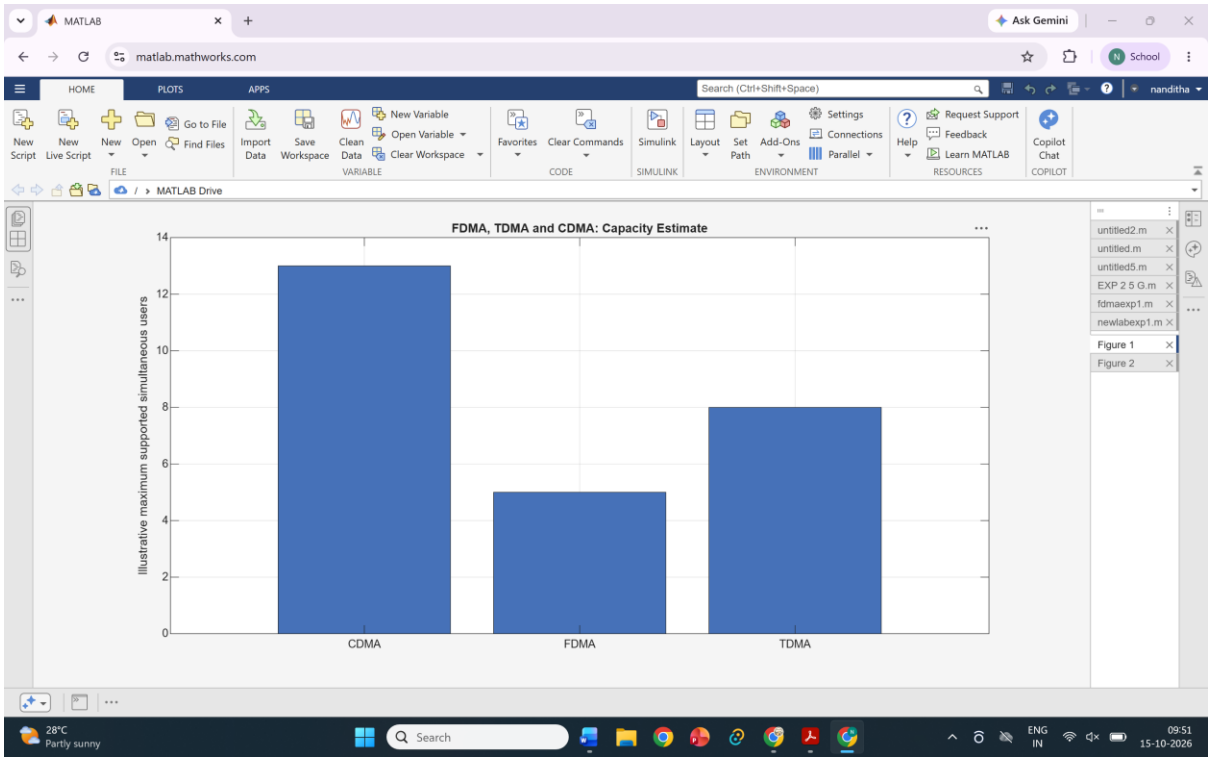
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
```

```
cdmaUsers, targetSIRdB);
% CDMA SIR as the number of simultaneous users increases
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
% Display a comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
results = table(method, supportedUsers);
disp(results);
% Plot estimated user capacity
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
% Plot estimated CDMA SIR as users are added
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

Result:

Maximum supported simultaneous users

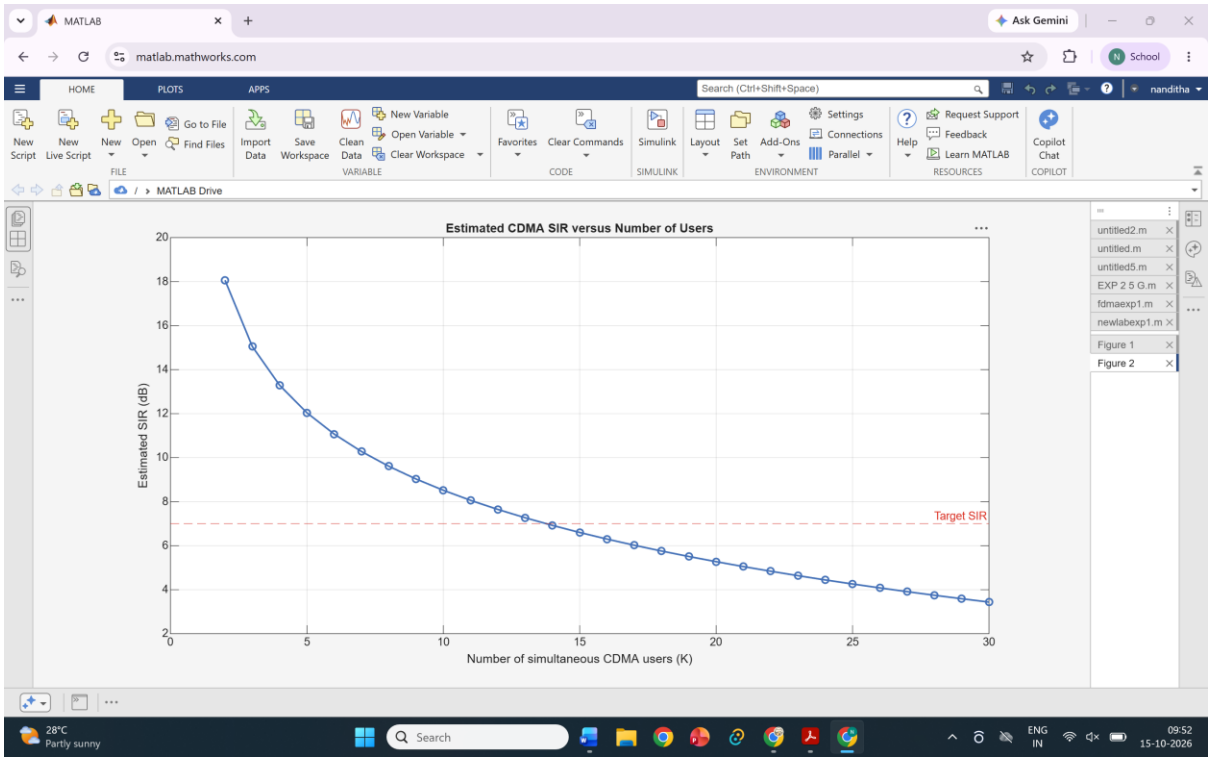




Objective 2: CDMA SIR analysis:

To analyze how the estimated CDMA signal-to-interference ratio changes as the number of simultaneous users increases.

Finding out the SIR for the technique with the highest number of users



Objective 3: Mobile-generation comparison:

To relate representative multiple-access techniques to the evolution of mobile communication

systems from 1G to 5G.

MATLAB Program:

```
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];

system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];

accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];

keyIdea = ["Separate frequency channels"; ...

"Separate time slots"; ...

"Code-based user separation"; ...

"Wideband code-based access"; ...

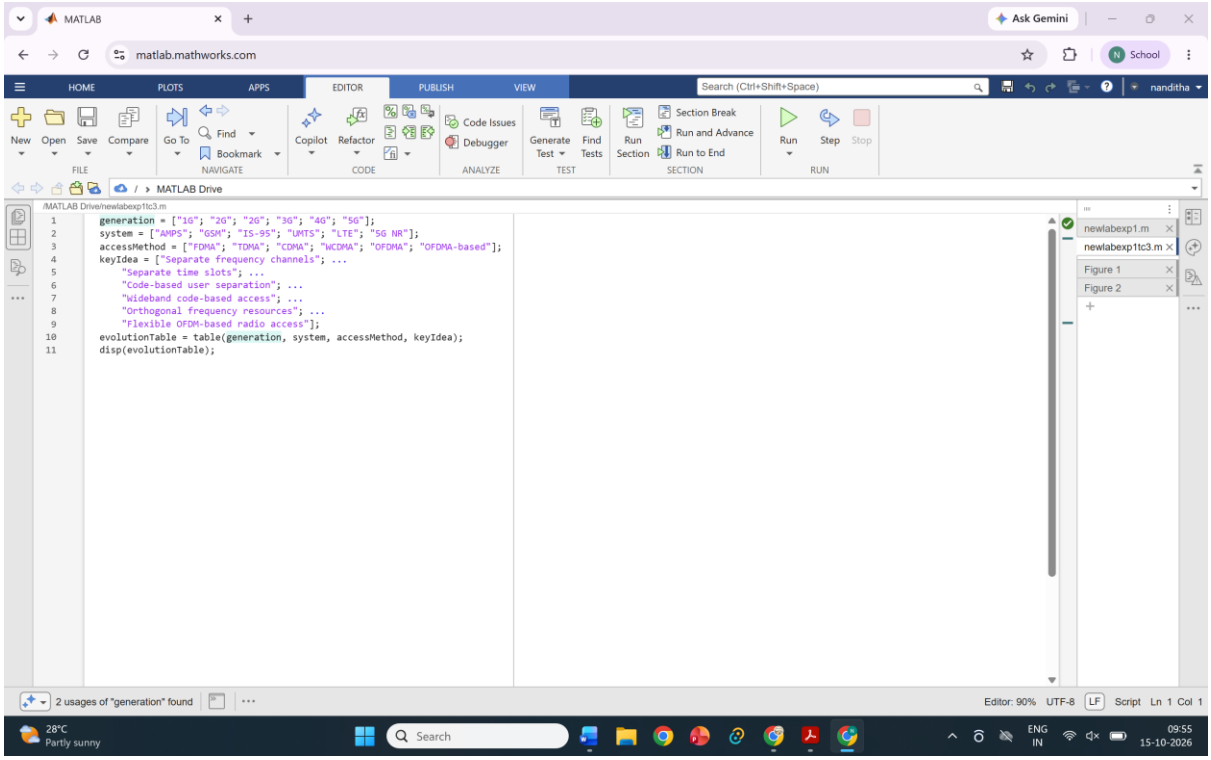
"Orthogonal frequency resources"; ...

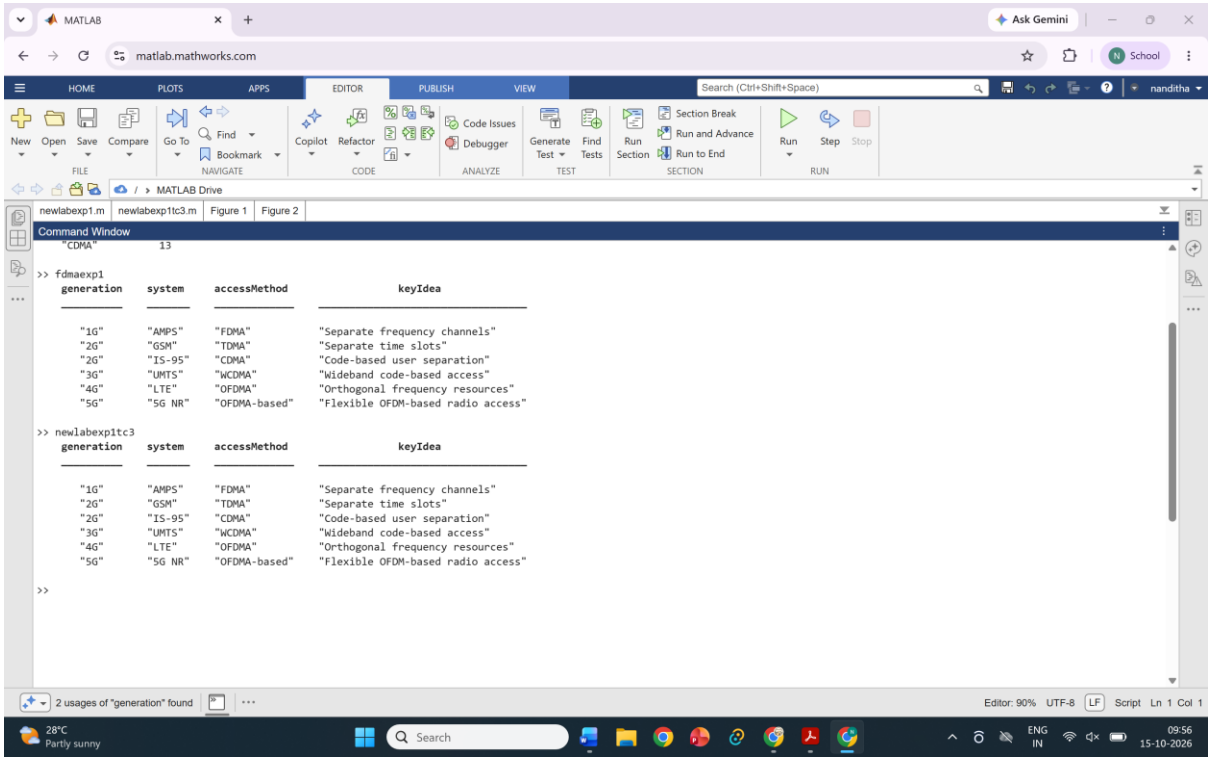
"Flexible OFDM-based radio access"];

evolutionTable = table(generation, system, accessMethod, keyIdea);

disp(evolutionTable);
```

Result:





Objective 4: Resource-allocation diagrams:

To visualize how FDMA, TDMA and CDMA assign frequency, time and code resources to

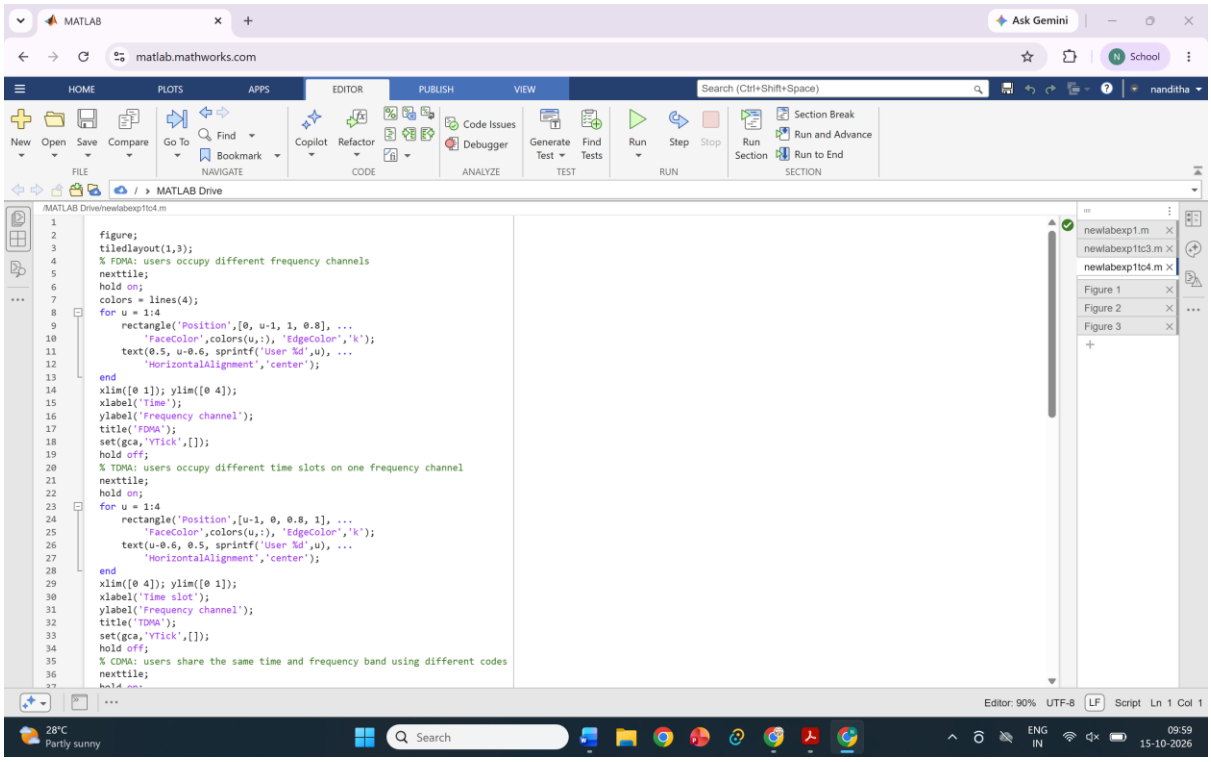
multiple users.

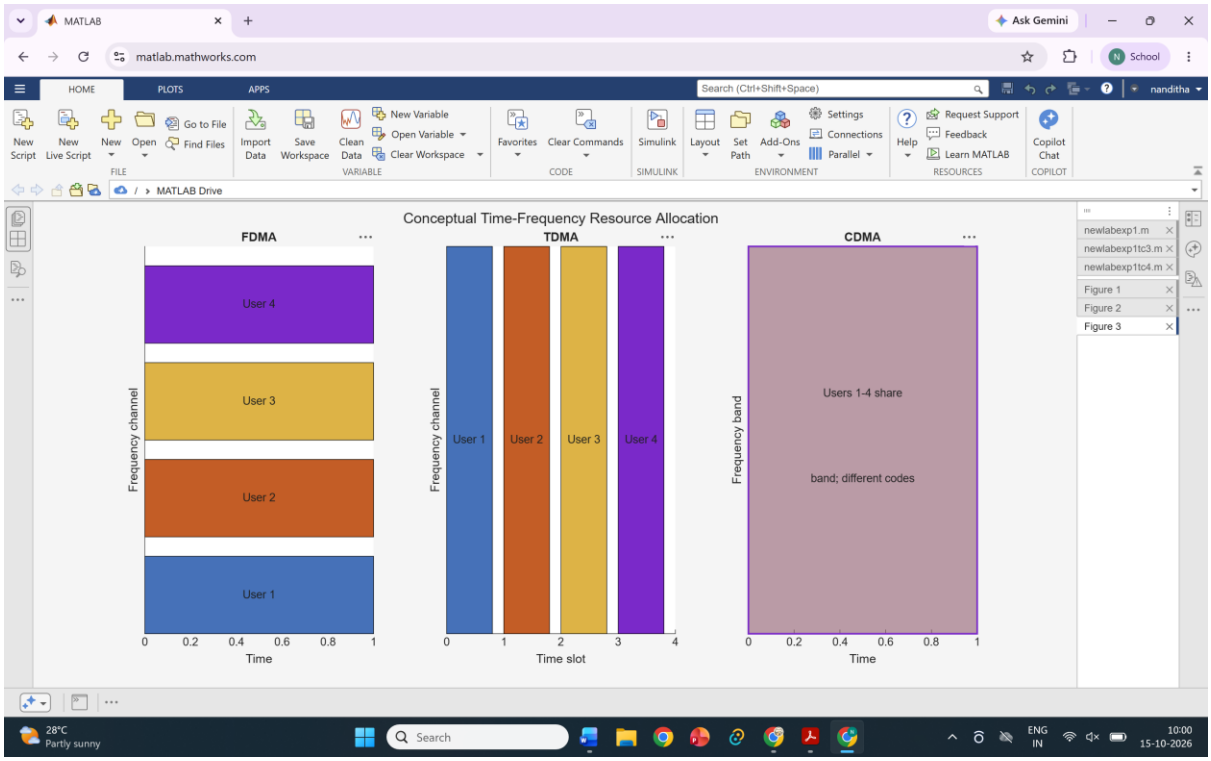
MATLAB Program:

```
Time-frequency resource-allocation diagrams
figure;
tiledlayout(1,3);
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
rectangle('Position',[0, u-1, 1, 0.8], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(0.5, u-0.6, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;
% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
rectangle('Position',[u-1, 0, 0.8, 1], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(u-0.6, 0.5, sprintf('User %d',u), ...
'HorizontalAlignment','center');
```

```
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
% CDMA: users share the same time and frequency band using different codes
nexttile;
hold on;
for u = 1:4
rectangle('Position',[0, 0, 1, 1], ...
'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
xlim([0 1]); ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca,'YTick',[]);
hold off;
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

Result :





Conclusion

The four MATLAB programs demonstrate how FDMA, TDMA and CDMA allocate communication resources and how the assumed parameters affect their estimated user capacity.

The CDMA analysis shows that estimated SIR decreases as more users share the same

frequency band. The mobile-generation table and resource-allocation diagrams connect these

techniques to the development of wireless communication systems.

These results are based on simplified assumptions. In particular, the model assumes ideal

FDMA and TDMA separation and does not include real-world effects such as fading, power

control, adjacent-channel interference or guard intervals. Therefore, the results illustrate the

operating principles; they should not be treated as a universal performance ranking of the three

methods.